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A Project Report

On

AI-Powered Micro-Internship Platform for Students and local Business

*A Project Report submitted in partial fulfillment of the requirements for the award of
the Degree of*

Master of Computer Applications

Specialization in Artificial Intelligence & Data Science

Submitted By

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CERTIFICATE

This is to Certify that the Project work entitled **Ai-Powered Micro-Internship Platform for Students and Local Business** is a bonafide work carried out by **Harshitha C** bearing the university seat number **24SUPMCAGL031** in partial fulfillment for the award of Master of Computer Applications of the Sapthagiri NPS University Bengaluru during the year 2025-26. It is certified that all corrections/suggestions indicated during the review process have been incorporated in the Project report. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the IV Semester Master of Computer Applications Degree.

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DECLARATION

I, **Harshitha C (24SUPMCAGL031)**, a student of IV Semester Master of Computer Applications (MCA), hereby solemnly declare that the project report titled **“Ai-Powered Micro-Internship Platform for Students and Local Business”** is a bonafide record of the original work carried out by me at the **School of Applied Science, Sapthagiri NPS University, Bengaluru**, during the **Academic year 2025–2026**.

I further declare that this project report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or any other academic qualification. I also declare that all sources of information used in this project have been duly acknowledged and referenced.

Place: Bengaluru

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ABSTRACT

In today's rapidly evolving educational and business environment, students require practical industry experience, while local businesses often need skilled individuals to complete short-term tasks and projects. Conventional internship platforms mainly focus on long-term internships and job opportunities, making it difficult for students to find flexible, skill-based micro-internships and for local businesses to identify suitable candidates for specific tasks. These limitations highlight the need for an intelligent platform that connects students with local businesses and provides opportunities to gain practical experience through short-term projects.

This project introduces **AI-Powered Micro-Internship Platform for Students and Local Businesses**, a web and mobile-based platform designed to connect students with local businesses through short-term, skill-oriented internship opportunities. The system provides a unified environment where students can create profiles, showcase their skills, browse available micro-internships, apply for suitable opportunities, and manage their applications. Local businesses can register, create internship opportunities, specify required skills and project details, and manage student applications.

The platform is developed using modern technologies including **React Native** for the frontend application, **Node.js and Express.js** for backend development, **MongoDB** for database management, and **Firebase** for authentication and supporting cloud services. The use of React Native enables the application to provide a consistent and responsive experience across mobile platforms. The MERN-based architecture provides efficient data management and scalable backend services, while Firebase supports secure user authentication and related application services.

The major modules of the system include Student Registration and Authentication, Business Registration and Authentication, Student Profile Management, Skill Management, Micro-Internship Creation, Internship Browsing and Search, Internship Application Management, Business Application Management, and Admin Management. Students can maintain their profiles with educational qualifications, technical skills, interests, and experience details. Businesses can publish micro-internship opportunities by providing project descriptions, required skills, duration, and other relevant information. Students can view available opportunities and submit applications based on their interests and skills.

The methodology begins with user registration and authentication, followed by profile creation and skill management. Local businesses can create and manage micro-internship opportunities, while students can browse and apply for suitable opportunities. Businesses can review submitted applications and select suitable students for their projects. The platform maintains application and internship information in MongoDB, enabling efficient management of users, opportunities, and application records. Firebase authentication provides secure login and user verification functionality.

The implementation demonstrates how a digital platform can effectively bridge the gap between students seeking practical experience and local businesses requiring short-term skilled support. The system reduces the difficulties involved in finding suitable micro-internship opportunities, simplifies the recruitment process for local businesses, and provides students with opportunities to develop practical skills and industry exposure. The student's contribution includes full-stack application development, React Native interface development, MERN stack integration, Firebase authentication, database management, internship management, application management, and administrative features.

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** provides a convenient, scalable, and user-friendly solution for connecting students and local businesses. By creating a digital ecosystem for short-term practical opportunities, the platform supports skill development, real-world learning, and collaboration between educational communities and local businesses.

Keywords: Artificial Intelligence, Micro-Internship, Students, Local Businesses, React Native, MERN Stack, MongoDB, Node.js, Express.js, Firebase, Skill Development, Internship Management

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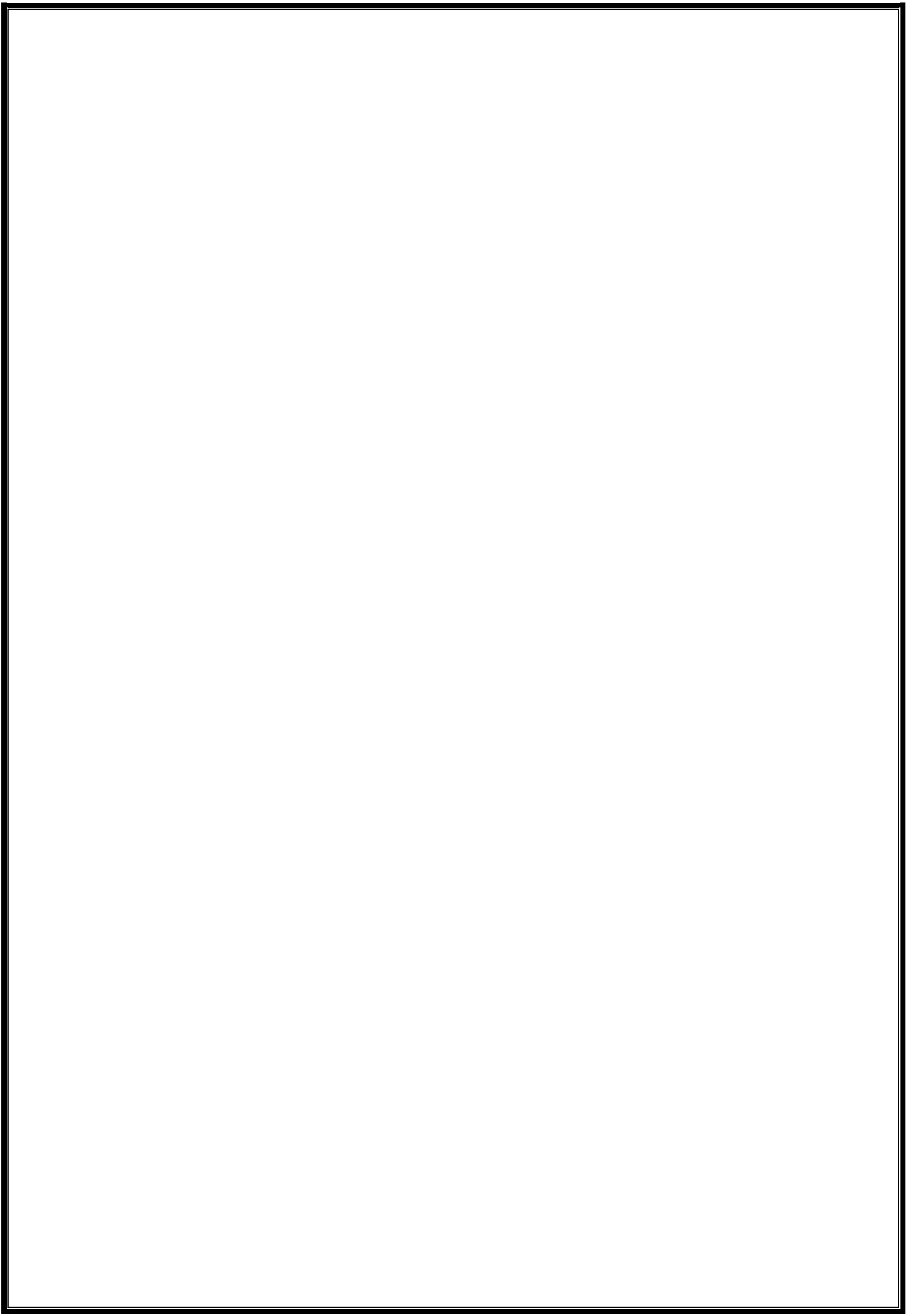
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CHAPTER 1. INTRODUCTION

1.1 Introduction

Digital technology has changed how students and employers connect. Students need practical experience, while local businesses need candidates for short-term projects. Existing job portals mainly focus on full-time jobs and long-term internships.

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** provides a centralized mobile application using **React Native**. Students can create profiles, upload resumes, search and apply for micro-internships, and track applications. Businesses can create profiles, post internships, and manage applications. **Firestore, and Firebase Storage** provide secure authentication, data management, and file storage.

1.2 Background

Micro-internships help students gain practical skills through short-term projects and allow local businesses to find suitable candidates without long-term commitments. However, existing platforms mainly support permanent jobs and traditional internships, making it difficult to find and manage micro-internships.

1.3 Problem Statement

Existing recruitment systems mainly focus on full-time jobs and long-term internships. Students find it difficult to find suitable micro-internships, while local businesses face challenges in posting opportunities and managing applications.

1.4 Objectives

- The primary objectives of the proposed project are:
- Develop a cross-platform micro-internship application using **React Native**.
- Provide secure authentication using **Firebase**.
- Allow students to create profiles and upload resumes.
- Allow businesses to create and manage internships.
- Enable students to search and apply for internships.
- Track application status.

- Provide an admin module for platform management.
- Simplify internship management and improve accessibility.

1.5 Scope of the Project

The platform is designed for students and local businesses. Students can create profiles, upload resumes, search and apply for internships, and track applications. Businesses can post internships, review applications, and manage recruitment through one mobile application.

1.6 Existing System

Existing internship and recruitment platforms mainly support full-time jobs and long-term internships. They provide limited support for short-term micro-internships and local businesses.

Limitations of the Existing System

- Focus on full-time jobs and long internships.
- Limited micro-internship support.
- Time-consuming registration and applications.
- Students need multiple platforms.
- Limited tools for local businesses.

1.7 Proposed System

The proposed platform provides a centralized mobile application for students and businesses. Students can create profiles, upload resumes, search and apply for internships, and track applications. Businesses can post internships, review applications, and manage recruitment. Firebase provides secure authentication, cloud storage, and data management.

Advantages of the Proposed System

- Centralized platform.
- Android and iOS support.
- Secure Firebase authentication.
- Easy internship posting and application.
- Resume upload.
- Application status tracking.

- User-friendly interface.
- Reduced manual work.
- Better communication and collaboration.

1.8 Feasibility Study

The feasibility of the proposed platform has been evaluated from technical, economic, operational, and schedule perspectives.

Technical Feasibility

React Native, JavaScript, and Firebase provide suitable technologies for secure and cross-platform development.

Economic Feasibility

The project uses open-source tools and Firebase services, reducing development and recruitment costs.

Operational Feasibility

The simple interface allows students, businesses, and administrators to manage their activities easily.

Schedule Feasibility

The project can be completed within the academic timeline using planned development phases.

1.9 Organization of the Report

The report is organized into the following chapters:

- **Chapter 1:** Introduction – Project overview and objectives.
- **Chapter 2:** Literature Survey – Related studies and technologies.
- **Chapter 3:** System Analysis and Design – Architecture, database, UML, and workflow.
- **Chapter 4:** System Implementation – Technologies and modules.
- **Chapter 5:** Results and Discussion – Outputs, testing, and performance.
- **Chapter 6:** Conclusion and Future Enhancements – Conclusion, limitations, and future improvements.

CHAPTER 2. LITERATURE SURVEY

2.1 Introduction

The literature survey reviews existing research and technologies related to internship management, mobile recruitment, cloud services, and cross-platform applications. It helps identify the limitations of current systems and the need for better solutions. Most existing platforms focus on full-time jobs or long-term internships, with limited support for micro-internships.

2.2 Background

Mobile and cloud technologies have made recruitment and internship management easier. Students can search for opportunities, create profiles, upload resumes, and apply online, while businesses can post opportunities and manage applications. However, existing platforms are mainly designed for traditional employment, creating a need for a platform focused on short-term opportunities.

2.3 Review of Existing Studies

Researchers have proposed several internship management and recruitment systems to improve communication between students and employers. Some of the important studies are summarized below.

Study 1: Online Internship Systems – Help institutions manage registrations and internship records but provide limited support for local businesses.

Study 2: Mobile Recruitment Applications – Platforms such as LinkedIn and Indeed support job searching and applications but mainly focus on regular employment.

Study 3: Cloud-Based Systems – Cloud platforms provide secure and accessible storage but are often designed for larger organizations.

Study 4: React Native – Supports Android and iOS development using a single codebase, reducing development effort.

Study 5: Firebase Applications – Provides authentication, cloud database, and storage services for secure application development.

Study 6: Digital Student Employment Platforms – Connect students with employers but generally focus on university placements or corporate recruitment.

Overall Summary: Existing technologies provide useful recruitment and internship features, but there is limited focus on connecting students with local businesses for micro-internships.

2.4 Comparative Analysis

Feature	Traditional Internship Systems	Online Recruitment Platforms	Proposed AI-Powered Micro-Internship Platform
Student Registration	Limited	Yes	Yes
Business Registration	No	Yes	Yes
Internship Posting	Limited	Yes	Yes
Internship Search	Limited	Yes	Yes
Resume Upload	No	Yes	Yes
Application Management	Manual	Partial	Yes
Application Status Tracking	No	Limited	Yes
Cross-Platform Mobile Support	No	Partial	Yes
Cloud-Based Data Storage	No	Partial	Yes
Admin Management	Limited	Partial	Yes
User-Friendly Mobile Interface	Basic	Good	Advanced
Centralized Internship Management	No	Partial	Yes

Table 2.4.1: Comparative Analysis

Comparison of traditional internship systems, online recruitment platforms, and the proposed AI-Powered Micro-Internship Platform based on key features and functionalities.

2.5 Research Gap Analysis

The main gaps in existing systems are:

- Limited support for micro-internships.
- Most platforms focus on full-time jobs or long-term internships.
- Limited support for local businesses.
- Students often use multiple platforms to find opportunities.
- Some systems have lengthy application processes.
- Limited dedicated mobile solutions.
- Internship activities are not fully centralized.

These gaps support the need for a simple and secure mobile platform connecting students with local businesses.

2.6 Motivation for the Proposed Work

The project aims to provide an easy platform where students can find and apply for short-term internships, while local businesses can post opportunities and manage applications. The proposed system reduces manual effort and brings these activities together in one mobile application.

2.7 Chapter Summary

This chapter reviewed existing internship, recruitment, mobile, and cloud-based systems. It identified their limitations and the need for a dedicated platform for student micro-internships and local businesses. The next chapter presents the **System Analysis and Design**, including architecture, database design, UML diagrams, and workflow.

CHAPTER 3. SYSTEM REQUIREMENTS

3.1 Introduction

System requirements define the hardware, software, technologies, and tools needed to develop and run the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. The system is developed using **React Native** and **Firebase** as the backend. It allows students to create profiles, upload resumes, search and apply for micro-internships, while businesses can register, post internships, and manage applications through a centralized mobile application.

3.2 Hardware Requirements

The proposed platform does not require specialized hardware for development or deployment. A standard desktop or laptop computer is sufficient for coding, testing, and debugging the application. Since Firebase services are cloud-based, a stable internet connection is essential for authentication, data synchronization, and file storage.

Recommended Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	4 GB or higher
Storage	SSD with at least 20 GB free space
Mobile Device	Android Smartphone (Testing)
Network	Broadband Internet Connection

Table 3.2 Hardware Requirements

Purpose of Hardware Components

- **Processor:** Executes the application, development tools, and compilation processes efficiently.
- **RAM:** Supports React Native development, Android Emulator, and multitasking during testing.
- **Storage:** Stores project files, application assets, source code, and development software.

- **Mobile Device:** Used to test application functionality on real Android devices.
- **Internet Connection:** Required for Firebase Authentication, Cloud Firestore synchronization, package installation, and cloud storage.

3.3 Software Requirements

The application is developed using modern open-source technologies that support secure, scalable, and maintainable mobile application development. These technologies provide backend services, cloud storage, authentication, and database management.

Software Requirements

Software	Purpose
Operating System	Windows 10/11
Programming Language	JavaScript
Mobile Framework	React Native
IDE	Visual Studio Code / Android Studio
Backend Services	Firebase
Database	Cloud Firestore
Authentication	Firebase Authentication
File Storage	Firebase Storage
Version Control	Git & GitHub
Testing	Android Emulator / Physical Device

Table 3.3 Software Requirements

React Native: Used to develop the cross-platform mobile application for Android and iOS.

JavaScript: Used for application logic, navigation, and UI functionality.

Firebase Authentication: Provides secure user registration and login.

Cloud Firestore: Stores user profiles, internships, and application data.

Firebase Storage: Stores resumes, profile images, and documents securely.

Android Studio: Used for Android testing, debugging, and emulator support.

Visual Studio Code: Used for coding, debugging, and development.

Git and GitHub: Used for version control and source code management.

3.4 Development Tools

Various software tools were utilized during the design, implementation, testing, and maintenance of the proposed platform.

Development Tools

Tool	Purpose
Visual Studio Code	Source Code Development
Android Studio	Emulator and Testing
Expo CLI	Running React Native Applications
Git	Version Control
GitHub	Source Code Repository
Firebase Console	Authentication, Database, and Storage Management

Table 3.4 Development Tools

Role of Development Tools

Visual Studio Code

Used for writing, editing, and debugging React Native and JavaScript source code.

Android Studio

Provides Android Emulator support for testing application functionality before deployment.

Expo CLI

Simplifies application development by enabling quick execution and testing of React Native projects.

Git and GitHub

Support version control, project backup, and collaborative software development.

Firebase Console

Used to configure authentication services, manage Firestore collections, monitor Firebase Storage, and maintain project settings.

3.5 Technology Stack

The proposed platform integrates frontend technologies, backend cloud services, database management, and development tools to provide a complete cross-platform mobile solution.

Technology Stack

Layer	Technologies Used
Frontend	React Native, JavaScript
Backend Services	Firebase
Authentication	Firebase Authentication
Database	Cloud Firestore
Cloud Storage	Firebase Storage
Development Tools	Android Studio, Visual Studio Code, Expo CLI
Version Control	Git, GitHub

Table 3.5 Technology Stack

Technology Stack Description

Frontend Layer: Provides the user interface for registration, dashboards, profiles, internships, and applications.

Technologies: React Native, JavaScript.

Backend Services Layer: Handles authentication, internship management, applications, and data synchronization.

Technologies: Firebase, Firebase Authentication, Cloud Firestore.

Database Layer: Stores user, profile, internship, and application data using Cloud Firestore.

Storage Layer: Securely stores resumes, profile images, and documents using Firebase Storage.

Version Control Layer: Manages project code and development history using Git and GitHub.

3.6 Software Architecture

The platform uses a **Three-Tier Architecture** consisting of Presentation, Application, and Database layers.

1. **Presentation Layer:** Provides the user interface for login, profiles, internship search, posting, and applications.

Technology: React Native, JavaScript.

2. **Application Layer:** Handles authentication, internship management, applications, and business logic.

Technology: JavaScript, Firebase Authentication, Cloud Firestore, Firebase Storage.

3. **Database Layer:** Stores user profiles, internships, applications, resumes, and status information securely.

Technology: Cloud Firestore, Firebase Storage.

3.6.1 Architecture Flow

The platform follows a structured workflow to manage internship activities efficiently.

Step 1: User Authentication: Students, businesses, and administrators register and log in using Firebase Authentication.

Step 2: Dashboard Access: Users access dashboards based on their roles.

Step 3: Profile Management: Students manage personal and academic details, while businesses manage organization profiles.

Step 4: Internship Posting: Businesses post internships with details such as title, skills, duration, and location.

Step 5: Internship Search: Students browse and search for suitable internship opportunities.

Step 6: Internship Application: Students apply for internships and submit their resumes.

Step 7: Resume Upload: Resumes are securely stored using Firebase Storage.

Step 8: Application Management: Businesses review applications and update status as Pending, Shortlisted, Selected, or Rejected.

Step 9: Database Synchronization: Cloud Firestore stores and synchronizes profiles, internships, applications, resumes, and status updates.

Step 10: Administrator Monitoring: Administrators monitor users, internships, applications, and platform activities.

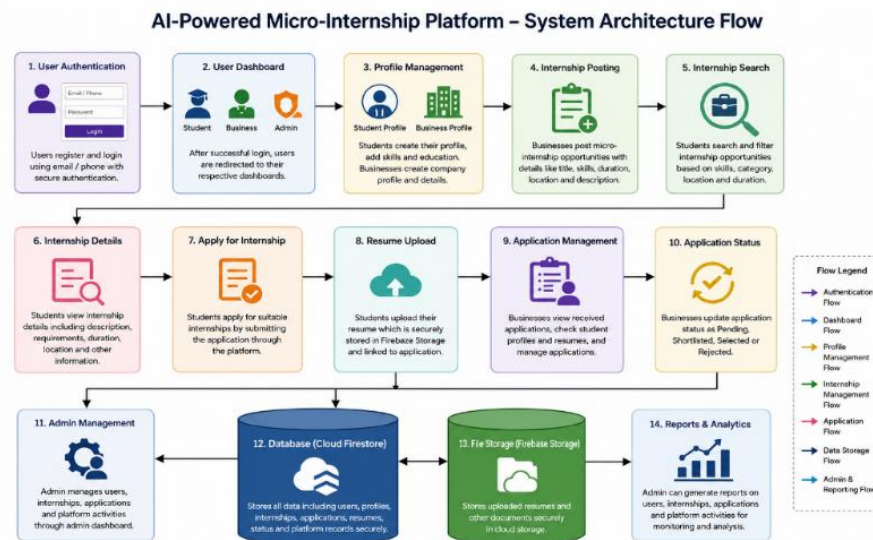


Fig:3.6.1 System Architecture Flow

3.7 Summary

This chapter presented the system requirements, technologies, development tools, and three-tier architecture of the AI-Powered Micro-Internship Platform for Students and Local Businesses. It also explained the main workflow, including authentication, profiles, internships, applications, resume storage, and admin management. These requirements provide a foundation for developing a secure and user-friendly mobile application.

CHAPTER 4. SOFTWARE REQUIREMENT SPECIFICATION

4.1 Introduction

The **Software Requirement Specification (SRS)** defines the functional and non-functional requirements of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. It describes the system features, user interactions, and requirements needed for development, testing, and maintenance.

The platform is developed using **React Native** and **Firebase**. Students can create profiles, upload resumes, search and apply for internships, while businesses can post internships, review applications, and manage recruitment.

4.2 Overall Description

The platform connects **students and local businesses** for short-term internship opportunities. Students can manage profiles, upload resumes, and apply for internships, while businesses can post internships and manage applications.

The system uses **React Native** for the mobile interface and **Firebase** for authentication, data management, and file storage.

4.3 User Characteristics

The platform has three main types of users:

1. Administrator

- Manage student and business accounts.
- Monitor internships and applications.
- Maintain platform security and activities.

2. Student

- Register and log in.
- Create and update profiles.
- Upload resumes.
- Search and apply for internships.
- Track application status.

3. Local Business

- Register and manage business profiles.
- Post internship opportunities.

- Review student applications.
- Update application status.

User Requirements: Users should have basic smartphone knowledge and a stable internet connection. The platform provides a simple and user-friendly interface for all users.

4.4 Functional Requirements

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** shall provide the following functional capabilities.

FR1 – User Authentication: Provides student, business, and admin login with secure Firebase Authentication and role-based access.

FR2 – Student Profile: Allows students to manage personal details, education, skills, and resumes.

FR3 – Business Profile: Allows businesses to create and update company information.

FR4 – Internship Management: Businesses can create, edit, delete, and manage internship postings.

FR5 – Internship Search: Students can search and view internships by title, category, or location.

FR6 – Application Management: Students can apply for internships and track their applications.

FR7 – Application Status: Businesses can review applications and update their status.

FR8 – Administrator Module: Admins can manage users, internships, applications, and platform activities.

4.5 Non-Functional Requirements

The proposed platform should satisfy the following quality requirements.

NFR1 – Performance: The system should provide fast loading, quick responses, and support multiple users.

NFR2 – Security: User data, authentication, and files should be securely protected with role-based access.

NFR3 – Reliability: The system should provide accurate data, stable performance, and reliable storage.

NFR4 – Usability: The platform should have a simple, user-friendly, and responsive interface.

NFR5 – Scalability: The system should support increasing users, internships, and applications.

NFR6 – Maintainability: The system should use modular and organized code for easy maintenance and future updates.

4.6 Use Case Diagram

The Use Case Diagram shows how the **Administrator, Student, and Local Business** interact with the platform. It represents key functions such as **login, profile management, internship management, applications, and user management**.

Actors

- **Administrator**
- **Student**
- **Local Business**

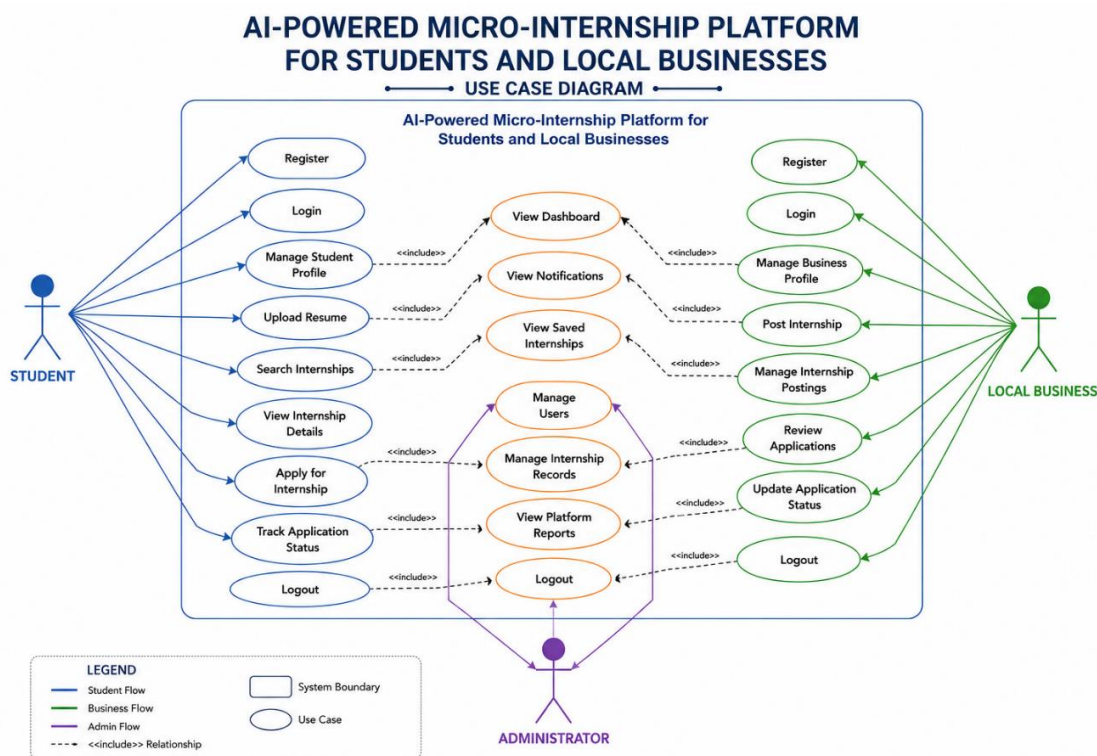


Fig 4.6 Use Case Diagram

4.7 Use Case Description

- 1. Register Student:** The student creates a new account by entering the required details. After successful registration, the student can log in to the platform.
- 2. Register Business:** The local business creates an account with business details. After registration, the business can access its dashboard.

- 3. Login:** Students, businesses, and administrators log in using their credentials. After successful login, they are directed to their respective dashboards.
- 4. Manage Student Profile:** Students can add and update their personal, education, and skill details. The updated information is saved in their profile.
- 5. Upload Resume:** Students can upload their resume to the platform. The resume is stored securely and can be used during applications.
- 6. Search Internships:** Students can browse and search for available internships. They can view the details of suitable internship opportunities.
- 7. Apply for Internship:** Students can select an internship and submit an application. The application is recorded for the business to review.
- 8. Post Internship:** Businesses can create and publish new internship opportunities. The posted internships become available for students to view.
- 9. Manage Internships:** Businesses can edit or delete their internship postings. This keeps the internship information updated.
- 10. Review Applications:** Businesses can view applications submitted by students. They can check student details and application information.
- 11. Update Application Status:** Businesses can update applications as Pending, Selected, or Rejected. The updated status is stored in the system.
- 12. Manage Users:** Administrators can manage student and business accounts. They can view, update, or remove user information when required.

4.8 External Interface Requirements

The platform provides interfaces for users, hardware, software, and communication.

User Interface: Provides login, registration, dashboards, profiles, internship search, applications, and resume upload through a simple mobile interface.

Hardware Interface: Supports Android smartphones, tablets, computers, and laptops with internet connectivity.

Software Interface: Uses React Native, JavaScript, Firebase, MongoDB, Android Studio, VS Code, and Expo for application development and data management.

Communication Interface: Uses secure HTTPS, Firebase Authentication, and cloud services for data communication and storage.

4.9 Constraints

- Stable internet is required.
- Only registered users can access features.
- Students need profiles before applying.
- Businesses need profiles before posting internships.
- Resume storage depends on Firebase capacity.
- Performance depends on network speed.
- Access is controlled by user roles.
- Device compatibility depends on supported Android/iOS versions.

4.10 Assumptions

The development and operation of the platform are based on the following assumptions:

- Users have basic smartphone knowledge.
- Students and businesses provide accurate information.
- Firebase services remain available.
- Users have reliable internet.
- Administrators monitor platform activities.
- Required software and tools are properly configured.
- Users follow platform policies.

4.11 Summary

This chapter presented the **SRS** of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**, covering interfaces, constraints, assumptions, and system requirements. These specifications provide a foundation for developing and testing a secure and user-friendly platform.

CHAPTER 5. SYSTEM DESIGN

5.1 Introduction

This chapter presents the design of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. It explains the overall system structure, module interactions, data flow, and database organization. The platform enables businesses to post micro-internship opportunities, allows students to apply for suitable roles, and supports efficient internship management.

The chapter also covers the system architecture, Data Flow Diagrams (DFDs), UML diagrams, database design, data dictionary, user interface design, functional design, and deployment diagram, providing a clear overview of the system implementation.

5.2 System Architecture

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** uses a three-tier architecture comprising the Presentation Layer, Application Layer, and Database Layer.

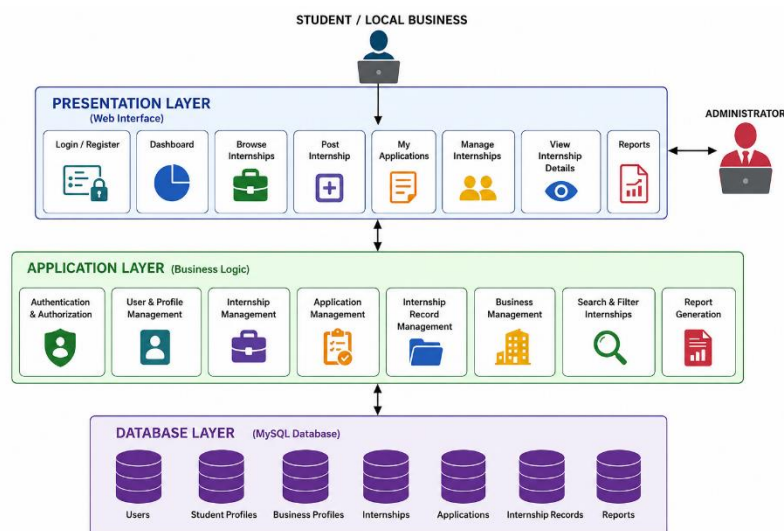


Fig 5.2 System Architecture

The platform adopts a three-tier architecture with the Presentation Layer, Application Layer, and Database Layer. The presentation layer allows students, local businesses, and administrators to access system features through a user-friendly interface. The application layer manages authentication, profiles, internship postings, applications, and internship tracking.

5.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) illustrates how information moves between users, system processes, and the database. It provides a clear representation of data processing within the platform.

5.3.1 DFD Level 0 (Context Diagram)

The DFD Level 0 (Context Diagram) presents the platform as a single process interacting with three external entities: **Student**, **Local Business**, and **Administrator**. Students register, log in, browse internships, submit applications, and check application status. Local businesses manage internship postings and review applications, while administrators oversee users, internship listings, and system records. The platform processes these requests and returns the appropriate information.

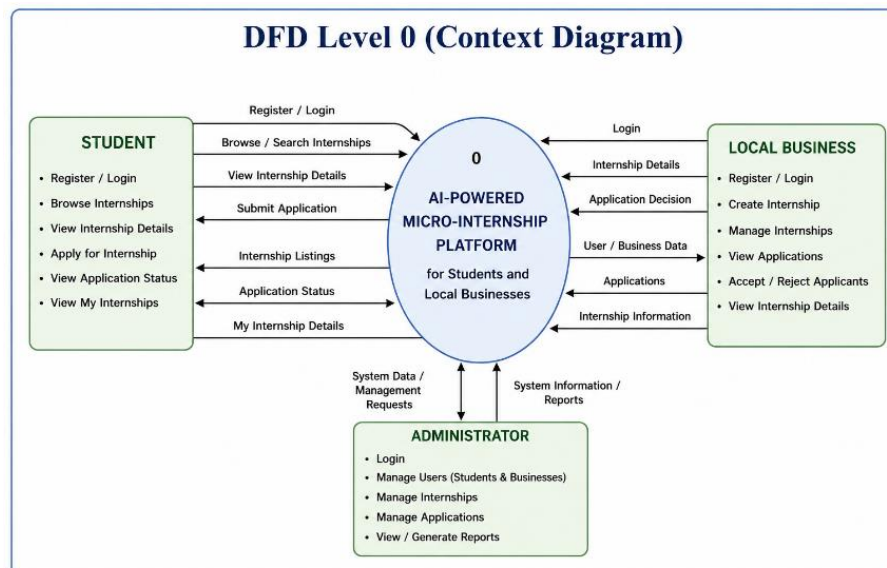


Fig 5.3.1 DFD Level 0

5.3.2 DFD Level 1

The DFD Level 1 divides the platform into key modules: **User Management**, **Student Profile Management**, **Business Profile Management**, **Internship Management**, **Application Management**, and **Internship Progress Management**. These modules process user requests and exchange data with the database to support internship posting, application, and progress tracking.

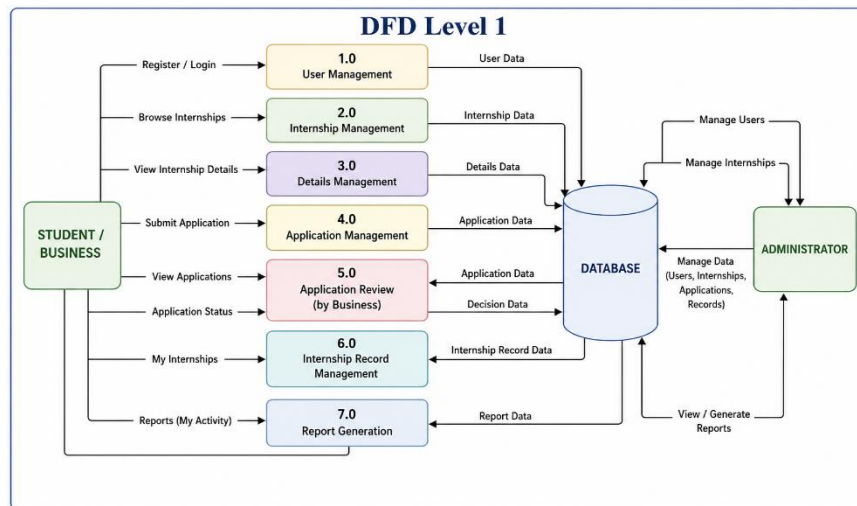


Fig 5.3.2 DFD Level 1

5.3.3 DFD Level 2

The DFD Level 2 provides a detailed view of the system processes, including user authentication, profile management, internship posting, application submission, application review, and internship tracking. Each process retrieves and stores information in the appropriate database to ensure efficient data management.

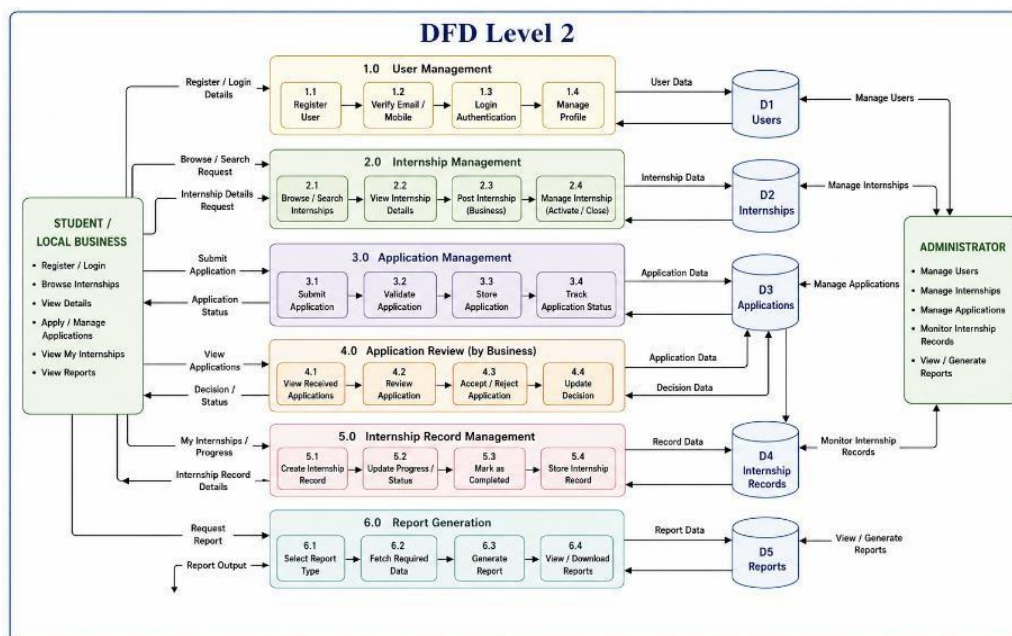


Fig 5.3.3 DFD Level 2

5.4 UML Diagrams

5.4.1 Sequence Diagram

The Sequence Diagram shows the interaction between the **Student, Local Business, Web Application, Application Service, and Database**. Businesses post internship opportunities, students submit applications, businesses review candidates, and the system updates application status for both users.

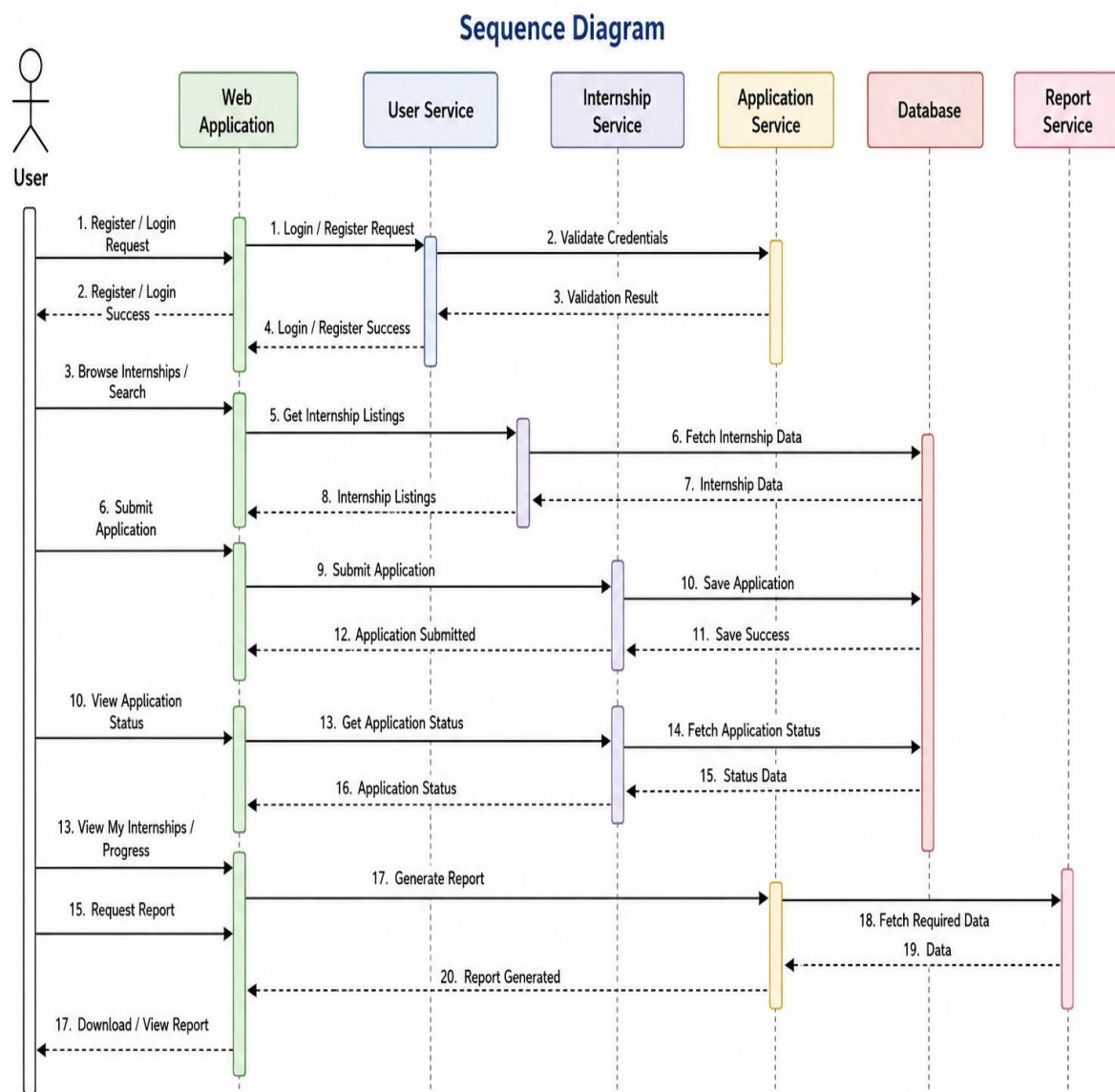


Fig 5.4.1 Sequence Diagram

5.4.2 Activity Diagram

The Activity Diagram illustrates the overall workflow of the platform. Users register or log in, businesses post internships, students apply for opportunities, businesses review applications, and accepted internships are tracked until completion.

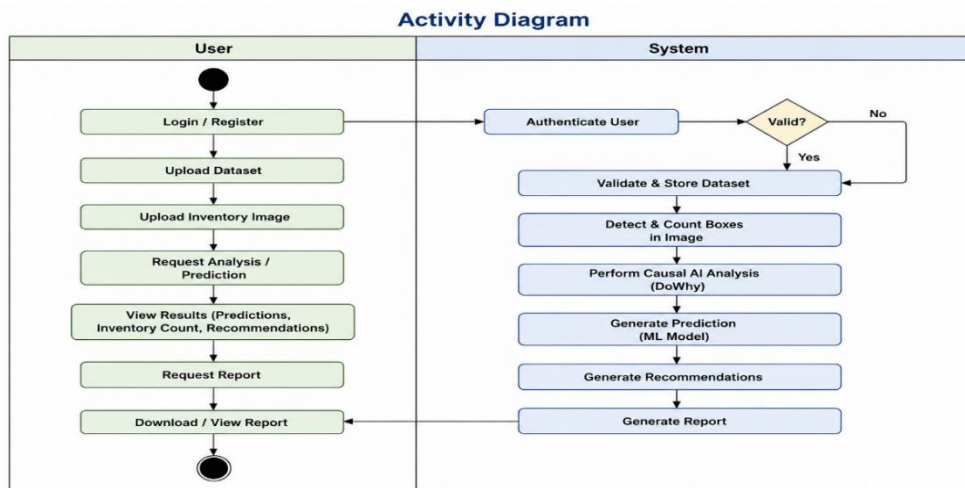


Fig 5.4.2 Activity Diagram

5.4.4 Class Diagram

The Class Diagram describes the main classes of the platform, including **Student**, **Business**, **Internship**, **Application**, **InternshipRecord**, and **Admin**. It shows the relationships between these classes and how they support internship posting, application management, record maintenance, and system administration.

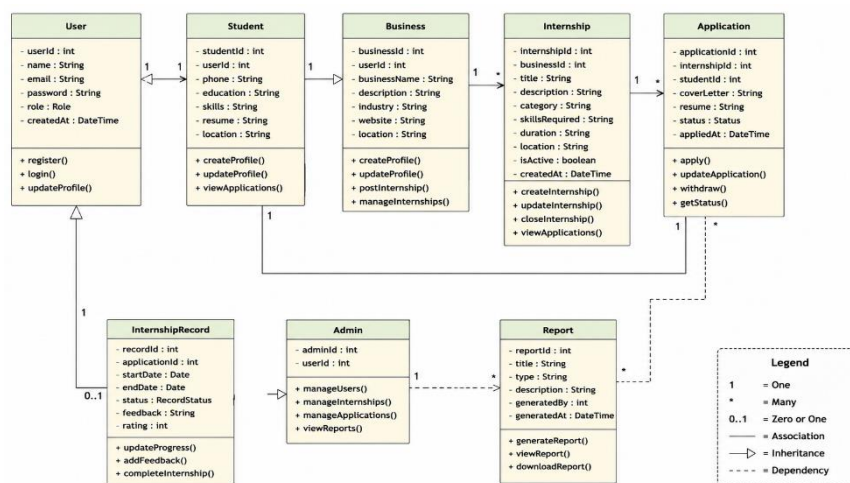


Fig 5.4.3 Class Diagram

5.5 Database Design

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** uses a relational database to store and manage user accounts, business profiles, internship postings, applications, internship records, and administrator data.

5.5.1 Entity Relationship (ER) Diagram

The Entity Relationship (ER) Diagram illustrates the relationships between the main entities: **Student**, **Business**, **Internship**, **Application**, **InternshipRecord**, and **Admin**. Students apply for internships posted by businesses, while applications connect students with specific internship opportunities. Internship records track accepted internships, and administrators manage users and system data. This design improves data integrity, minimizes redundancy, and supports efficient data management.

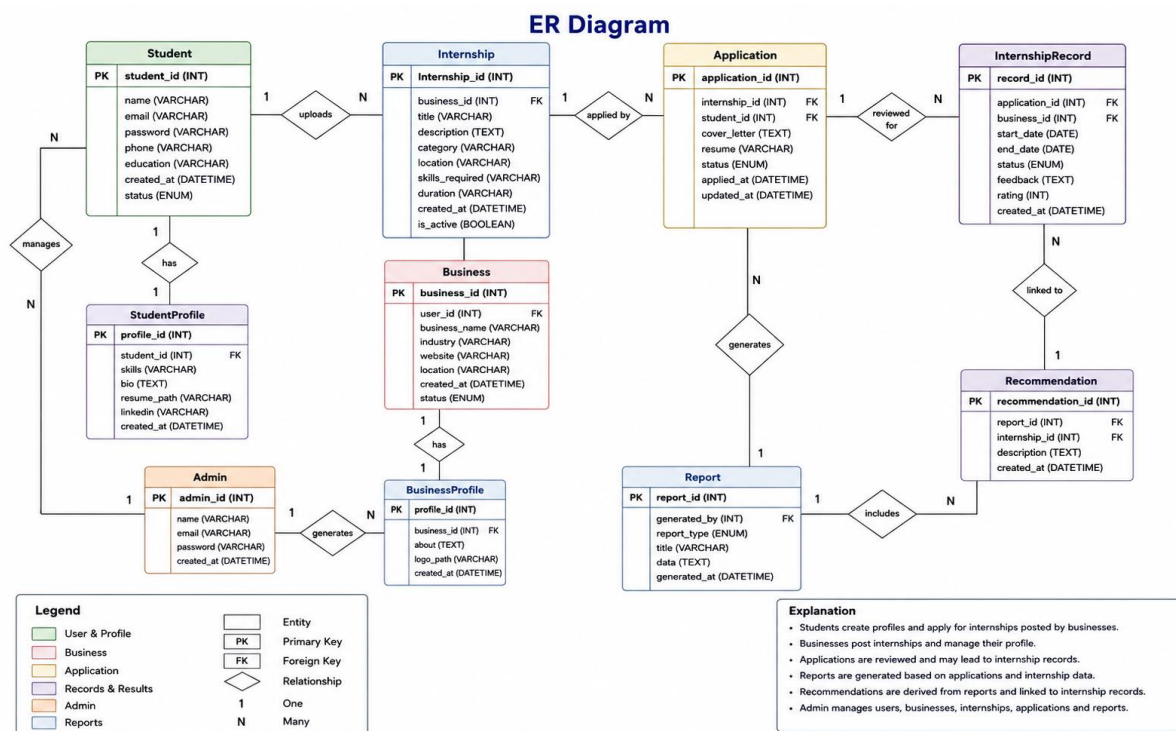


Fig 5.5.1 Entity Relationship (ER) Diagram

5.6 Data Dictionary

The Data Dictionary describes the database fields used in the platform, including each field's name, data type, key, and purpose. It serves as a reference for maintaining a consistent and well-structured database.

5.6.1 Data Dictionary Tables

Table 5.6.1 User

Field Name	Data Type	Description
User_ID	String	Unique ID of the user
Name	String	Name of the user
Email	String	Registered email address
Password	String	User login password
Role	String	User role: Student, Business, or Administrator
Phone	String	User contact number
Created_At	Timestamp	Date and time of account creation
Status	String	Current account status

Table 5.6.1 User

Indicates whether the user account is active, inactive, or blocked. It helps administrators monitor and manage the current account status.

Table 5.6.2 Student

Field Name	Data Type	Key	Description
student_id	INT	PK	Unique student ID
user_id	INT	FK	User ID
phone	VARCHAR(20)	-	Contact number
education	VARCHAR(100)	-	Education qualification
skills	TEXT	-	Student skills
location	VARCHAR(100)	-	Student location

Table 5.6.2 Student

This table is the Student Profile table. It stores the personal, educational, and professional information of students registered in the system.

Table 5.6.3 Business

Field Name	Data Type	Key	Description
business_id	INT	PK	Unique business ID
user_id	INT	FK	User ID
business_name	VARCHAR(150)	-	Business name
industry	VARCHAR(100)	-	Business category
location	VARCHAR(100)	-	Business location
description	TEXT	-	Business description

Table 5.6.3 Business

The **Business table** stores the details of businesses registered in the system. It contains information such as business name, industry, location, and description.

Table 5.6.4 Internship

Field Name	Data Type	Key	Description
internship_id	INT	PK	Unique internship ID
business_id	INT	FK	Business ID
title	VARCHAR(150)	-	Internship title
description	TEXT	-	Internship description
category	VARCHAR(100)	-	Internship category
duration	VARCHAR(50)	-	Internship duration
location	VARCHAR(100)	-	Internship location
created_at	DATETIME	-	Posting date

Table 5.6.4 Internship

The Internship table stores details of internship opportunities posted by businesses. It includes the internship title, description, category, duration, location, and posting date.

Table 5.6.5 Application

Field Name	Data Type	Key	Description
application_id	INT	PK	Unique application ID
internship_id	INT	FK	Internship ID
student_id	INT	FK	Student ID
resume	VARCHAR(255)	-	Resume file
cover_letter	TEXT	-	Cover letter
status	VARCHAR(20)	-	Application status
applied_at	DATETIME	-	Application date

Table 5.6.5 Application

The Application table stores information about internship applications submitted by students. It contains the internship and student details, along with the resume, cover letter, application status, and application date.

Table 5.6.6 Internship Record

Field Name	Data Type	Key	Description
record_id	INT	PK	Internship record ID
application_id	INT	FK	Application ID
start_date	DATE	-	Internship start date
end_date	DATE	-	Internship end date
status	VARCHAR(20)	-	Internship status
feedback	TEXT	-	Internship feedback

Table 5.6.6 Internship Record

The Internship Record table stores the details of a student’s internship after an application is accepted. It records the internship start and end dates, current status, and feedback.

Table 5.6.7 Report

Field Name	Data Type	Key	Description
report_id	INT	PK	Unique report ID
generated_by	INT	FK	User/Admin ID
report_type	VARCHAR(50)	-	Report category
generated_at	DATETIME	-	Report generation date

Table 5.6.7 Report

The Report table stores information about reports generated by users or administrators. It includes the report type and the date and time when the report was generated.

Table 5.6.8 Admin

Field Name	Data Type	Key	Description
admin_id	INT	PK	Unique administrator ID
name	VARCHAR(100)	-	Administrator name
email	VARCHAR(100)	-	Administrator email
password	VARCHAR(255)	-	Administrator password

Table 5.6.8 Admin

The **Admin table** stores the login and identification details of system administrators. It contains the administrator’s name, email, and password. The `admin_id` uniquely identifies each administrator and helps manage administrative activities within the system.

The data dictionary defines the main database tables used in the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. It includes user accounts, student and business profiles, internship postings, applications, internship records, reports, and

administrator data. Primary and foreign keys establish relationships between tables, supporting efficient and reliable data management.

5.7 Functional Design

The functional design describes the major modules of the platform and their responsibilities. These modules work together to provide secure access, internship management, application processing, and system administration.

5.7.1 User Authentication Module - Handles user registration, login, password management, and role-based access.

5.7.2 Student Profile Management Module - Allows students to create profiles, upload resumes, and manage their information.

5.7.3 Business Profile Management Module - Enables businesses to create and update their company details.

5.7.4 Internship Management Module - Allows businesses to create, update, delete, and manage internship opportunities.

5.7.5 Application Management Module - Handles internship applications, resume submission, and application status.

5.7.6 Internship Record Management Module - Tracks accepted internships, progress, status, completion, and feedback.

5.7.7 Report Generation Module - Generates and manages reports related to users, applications, and internships.

5.7.8 Admin Module - Provides administrators with control over users, internships, applications, records, reports, and system activities.

Overall: The functional design divides the platform into modules for secure access, profile management, internship handling, applications, tracking, reporting, and administration.

5.8 User Interface Design

The **AI-Powered Micro-Internship Platform** provides a simple and user-friendly interface developed using **React Native and MERN technologies**. It helps students, businesses, and administrators manage internships and applications efficiently.

5.8.1 Login Interface

Allows users to securely log in using their email and password.

5.8.2 Registration Interface

Allows students and businesses to create accounts by entering basic details.

5.8.3 Dashboard Interface

Displays important information such as internships, applications, and quick navigation options.

5.8.4 Student Profile Interface

Allows students to manage their personal details, education, skills, and resume.

5.8.5 Business Profile Interface

Allows businesses to manage their organization and contact details.

5.8.6 Internship Management Interface

Enables businesses to create, edit, delete, and manage internship postings.

5.8.7 Application Interface

Allows students to apply for internships and businesses to review applications.

5.8.8 Internship Record Interface

Maintains internship details, status, feedback, and completion records.

5.8.9 Reports Interface

Displays basic internship and application records for monitoring.

5.8.10 Admin Interface

Allows administrators to manage users, internships, applications, and overall platform activities.

5.9 Deployment Diagram

The **Deployment Diagram** illustrates how the **AI-Powered Micro-Internship Platform for Students and Local Businesses** is deployed and how its components communicate. The platform follows a **three-tier client-server architecture** consisting of client devices, a web server, an application server, and a MySQL database.

Users access the platform through a web browser on desktops, laptops, tablets, or mobile devices. The web server receives user requests and forwards them to the application server, which processes authentication, profile management, internship postings, applications, and report generation. The application server interacts with the MySQL database to store and retrieve data before sending the results back to users. This architecture provides secure, reliable, and efficient system operation.

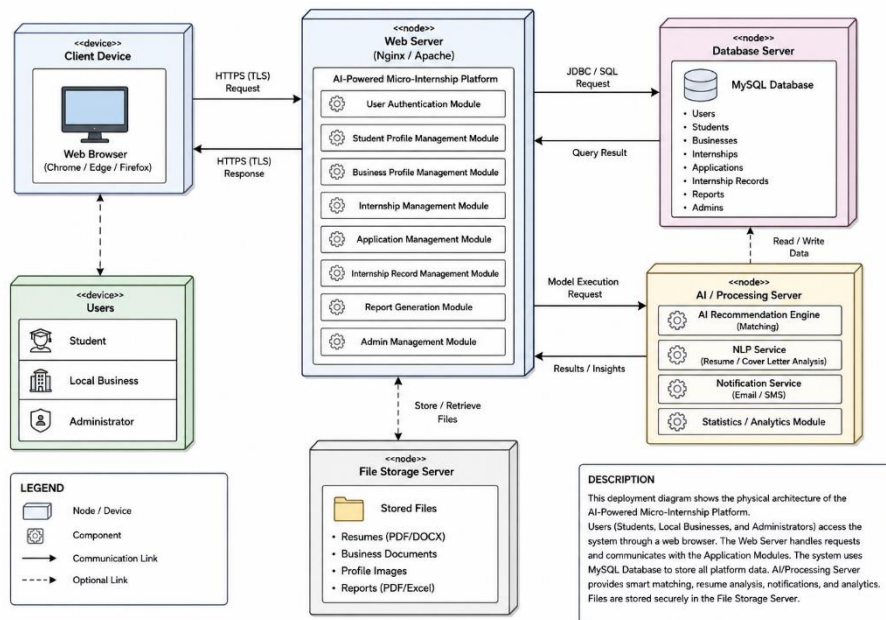


Fig 5.9 Deployment Diagram

5.10 Summary

This chapter presented the system design of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. It covered the system architecture, DFDs, UML diagrams, database design, data dictionary, functional design, user interface design, and deployment diagram. Together, these components describe the system structure, data flow, and module interactions, providing a foundation for implementing a secure, reliable, and efficient internship management platform.

CHAPTER 6. IMPLEMENTATION

6.1 Introduction

This chapter describes the implementation of the AI-Powered Micro-Internship Platform for Students and Local Businesses. The platform is developed using Spring Boot for the backend, React.js, HTML5, CSS3, Bootstrap, and JavaScript for the frontend, and MySQL for data storage. Spring Security, JWT Authentication, Hibernate (JPA), and REST APIs are used to provide secure authentication, efficient data management, and communication between the client and server.

6.2 Module-wise Implementation

The platform is divided into independent modules, each responsible for a specific function.

6.2.1 User Authentication Module - Handles user registration and secure login with role-based access.

6.2.2 Dashboard Module - Displays internships, applications, and important user activities.

6.2.3 Student Profile Module - Manages student details, education, skills, and resume.

6.2.4 Business Profile Module - Manages business details and contact information.

6.2.5 Internship Management Module - Allows businesses to create, update, delete, and manage internships.

6.2.6 Application Management Module - Allows students to apply for internships and businesses to review applications.

6.2.7 Internship Record Module - Maintains internship progress, status, and completion records.

6.2.8 Report Module - Provides basic internship and application reports.

6.2.9 Admin Module - Manages users, internships, applications, and overall platform activities.

6.3 Class-wise Implementation (OOP-Based)

The platform follows an object-oriented design where each class performs a specific responsibility.

Class	Responsibility
User	Authentication and user management
Student	Student profile management
Business	Business profile management
Internship	Internship management
Application	Application processing
InternshipRecord	Internship tracking
Report	Report generation
Admin	Platform administration

Table 6.3 Class-wise Implementation

It describes the main classes of the system and their respective responsibilities. Each class handles a specific function, such as user management, student and business profiles, internships, applications, internship tracking, reports, and administration.

6.4 Algorithms / Pseudocode

Algorithm 1: User Login

- Enter email and password.
- Validate credentials.
- If valid, open dashboard; otherwise show error.

Algorithm 2: Student Registration

- Enter registration details.
- Validate and save the details.
- Display registration success.

Algorithm 3: Internship Posting

- Business enters internship details.

- Validate the information.
- Save and publish the internship.

Algorithm 4: Internship Application

- Student selects an internship.
- Upload resume and submit application.
- Save the application details.

Algorithm 5: Application Review

- Business views applications.
- Accept or reject an application.
- Update and display the application status.

6.5 Database Implementation

The platform uses **MySQL** as the database, with **Hibernate (JPA)** managing communication between the application and the database. The main tables include **User, Student, Business, Internship, Application, InternshipRecord, Report, and Admin**. Primary and foreign keys maintain data integrity, while CRUD operations support efficient data management.

6.6 User Interface Implementation

The interface is developed using **React Native** with a simple and responsive design.

Implemented Interfaces:

- Login and Registration
- Dashboard
- Student and Business Profiles
- Internship Management
- Application Management
- Admin Dashboard

6.7 Module Integration

All modules are integrated into a single workflow, allowing users to register, post internships, apply for opportunities, manage applications, and track internship records.

6.8 Security and Validation

The system provides:

- Firebase Authentication
- Role-Based Access
- Input Validation
- Secure Data Storage
- Resume File Validation

6.9 Performance Optimization

Performance is improved using:

- Efficient MongoDB Queries
- REST APIs
- Optimized React Native Components
- Reduced Database Operations
- Responsive UI

6.10 Summary

This chapter explained the implementation of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. It covered the system modules, classes, database, user interface, security, and performance. These features make the platform secure and efficient for managing internships. The next chapter presents the testing and evaluation of the system.

CHAPTER 7. TESTING

7.1 Introduction

Testing was performed to ensure the **AI-Powered Micro-Internship Platform for Students and Local Businesses** works correctly, securely, and reliably. Major modules such as **authentication, profiles, internships, applications, and administration** were tested.

7.2 Testing Strategy

The application was tested using the following methods:

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Performance Testing
- Security Testing

7.3 Unit Testing

Individual modules such as Authentication, Profiles, Internship Management, Applications, Reports, and Admin were tested separately.

Result: All modules worked as expected.

7.4 Integration Testing

Integration testing checked the connection between login, profiles, internships, applications, reports, and database operations.

Result: Data was transferred correctly between modules.

7.5 System Testing

The complete application was tested to verify authentication, profiles, internships, applications, reports, and admin functions.

Result: The system met the required functional requirements.

7.6 User Acceptance Testing (UAT)

UAT verified registration, profiles, internship search and posting, applications, status tracking, and dashboard navigation.

Result: Users found the application simple and suitable for managing micro-internships.

7.7 Test Cases

Test Case	Scenario	Expected Result	Status
TC01	User Registration	Account created successfully	Pass
TC02	User Login	Login successful	Pass
TC03	Student Profile	Profile saved successfully	Pass
TC04	Business Profile	Profile saved successfully	Pass
TC05	Internship Posting	Internship published successfully	Pass
TC06	Internship Search	Matching internships displayed	Pass
TC07	Internship Application	Application submitted successfully	Pass
TC08	Application Status	Correct status displayed	Pass
TC09	Report Generation	Report generated successfully	Pass
TC10	Admin Login	Dashboard displayed	Pass

Table 7.7 Test Cases

Test Cases presents the test cases used to verify the major functions of the system. All test cases passed successfully, confirming that the platform functions as expected.

7.8 Test Results

Testing confirmed that all major modules operated correctly. Authentication, internship management, application processing, reporting, and administrative functions produced the expected results. The application remained stable during testing.

7.9 Performance Testing

Performance testing measured response time and system efficiency for key operations such as login, dashboard loading, internship search, and report generation.

Result: The application responded quickly under normal usage conditions, and database operations were completed efficiently.

7.10 Security Testing

Security testing verified authentication, password protection, role-based access, session management, input validation, and secure file uploads. The system successfully protected user data and prevented unauthorized access.

7.11 Summary

This chapter presented the testing activities performed on the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. Different testing methods confirmed that the application is functional, secure, reliable, and performs efficiently. The successful test results indicate that the platform is ready for deployment and practical use.

CHAPTER 8. RESULTS AND DISCUSSION

8.1 Experimental Results

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** was successfully implemented and evaluated using student, business, and administrator accounts. Testing confirmed that the core features functioned correctly, including user registration, login, profile management, internship posting, internship search, application submission, resume upload, and application status tracking.

8.2 Output Screens

This section shows the main application screens, including login, dashboards, profiles, internships, applications, and admin functions.

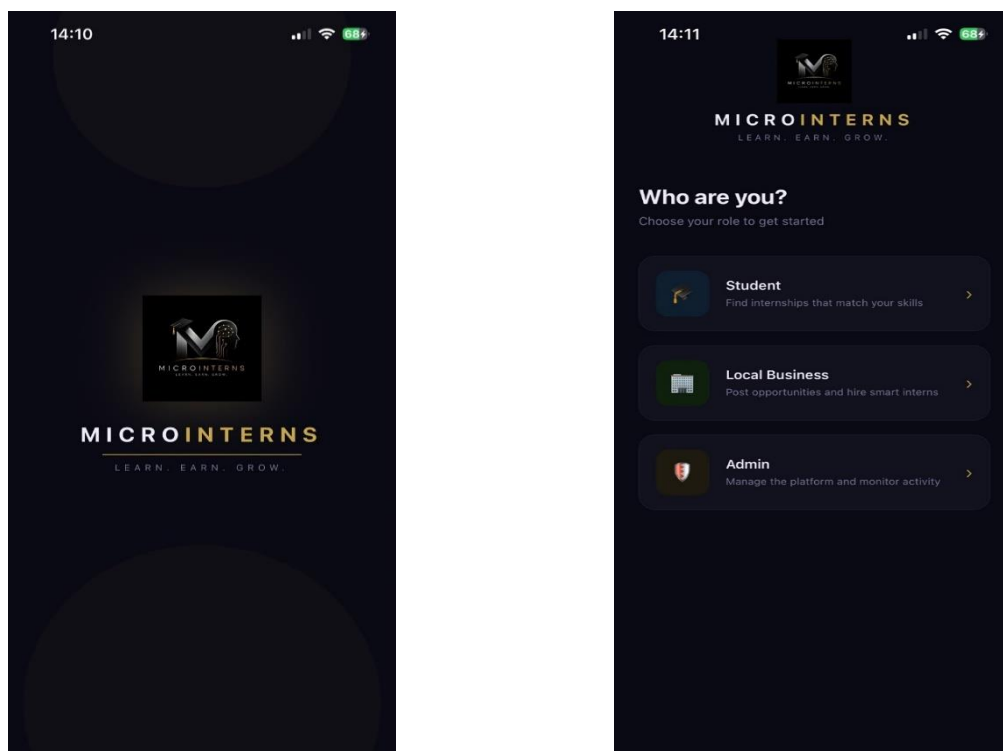


Fig 8.2.1 – Splash Screen and onboarding screen

The **MicroInterns** app welcomes users with the “**Learn. Earn. Grow.**” splash screen and then allows them to select their role as **Student, Local Business, or Admin** to access the appropriate platform features.

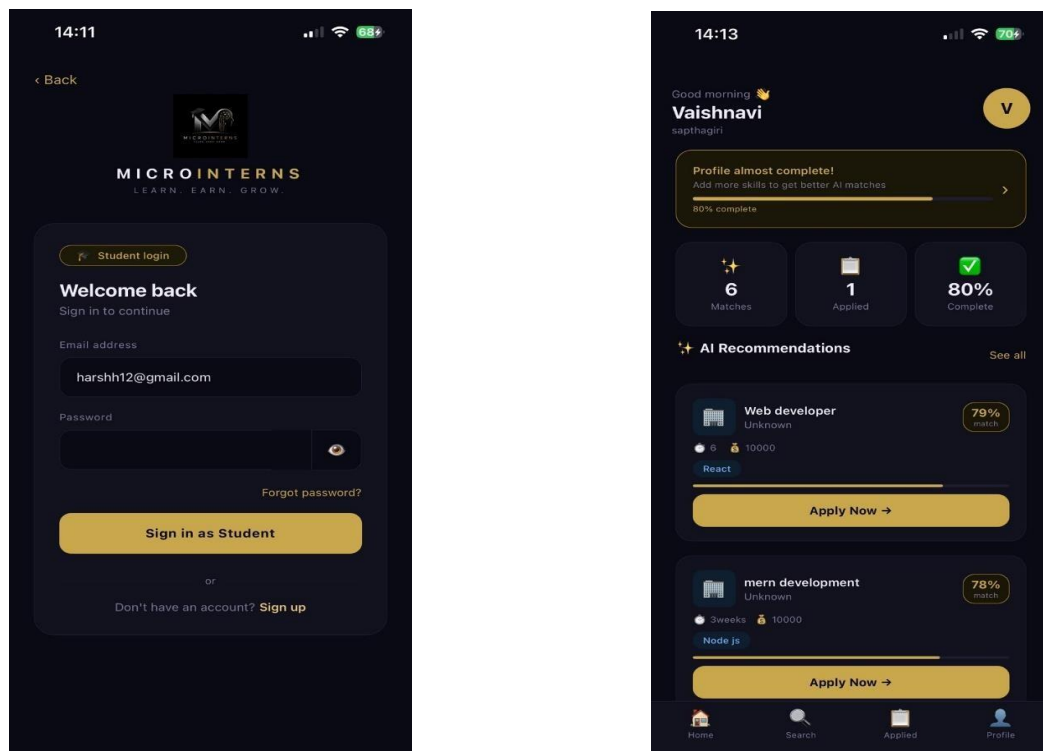


Fig 8.2.2 – Signup screen & Student Dashboard

Shows the signup screen where students create their accounts. The student dashboard provides access to internships, applications, and profile features.

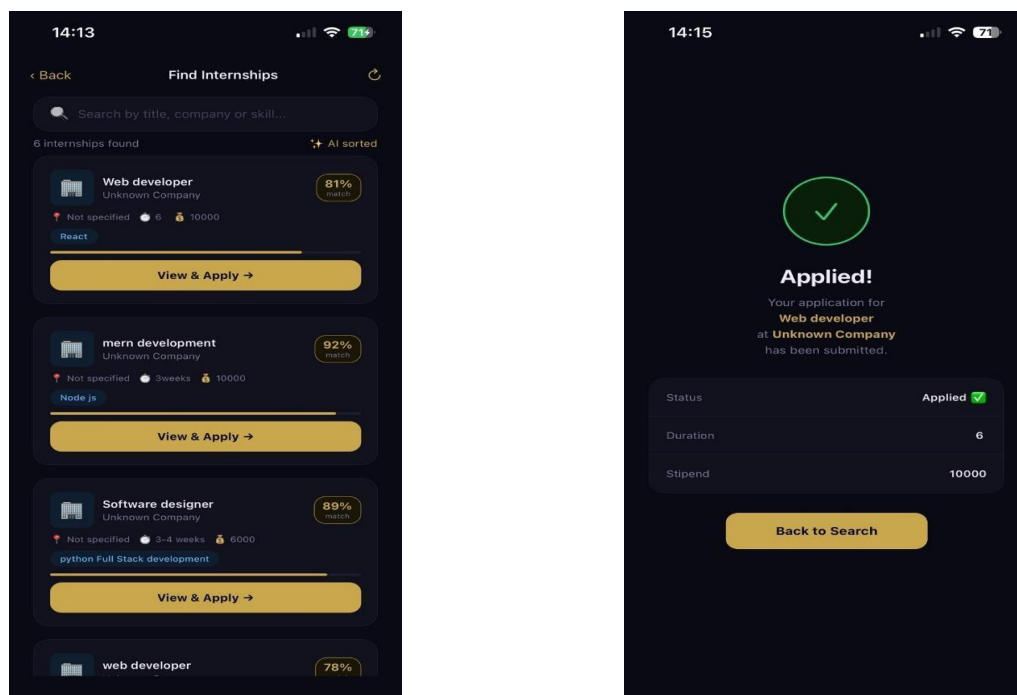


Fig 8.2.3 – Search Internship screen & Apply screen

Shows the internship search screen for finding suitable opportunities. The apply screen allows students to submit applications for selected internships.

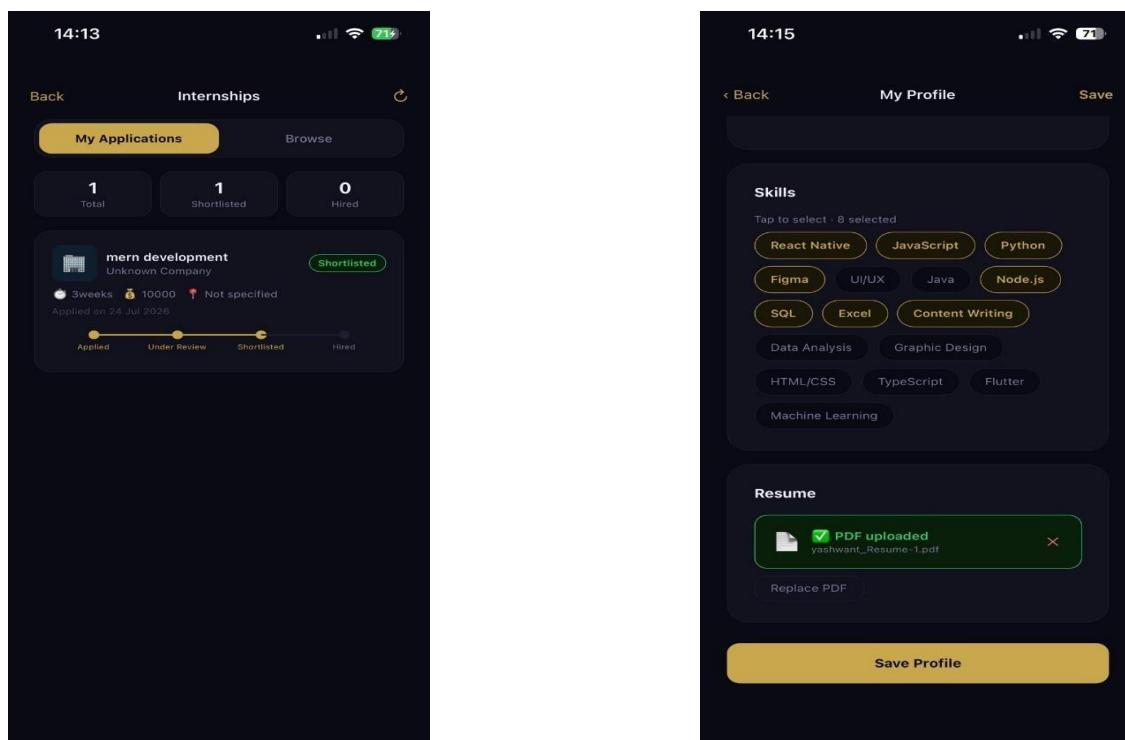


Fig 8.2.4 – Track Application screen & Profile jscreen

Shows the application tracking screen for checking application status. The profile screen allows students to view and update their personal details.

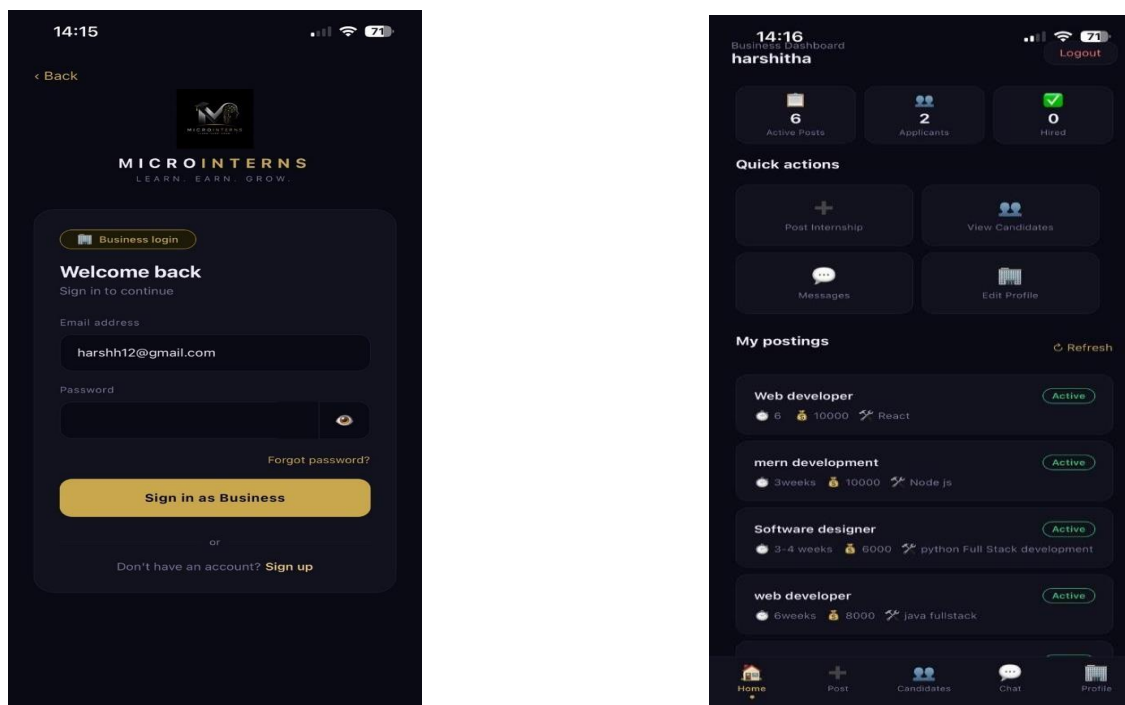


Fig 8.2.5 – Buisness SignIn & Dashboard Screen

Shows the business sign-in screen for secure login. The dashboard provides businesses with options to manage their activities and internships.

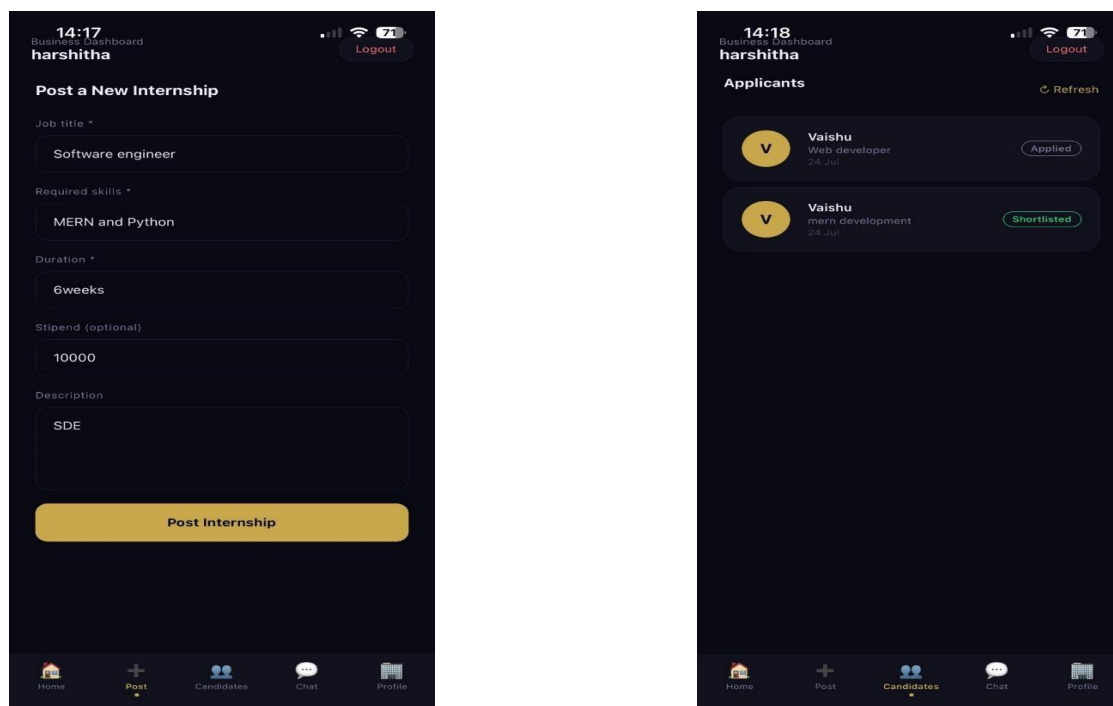


Fig 8.2.6 – Posting Internship & Applied candidate Screen

Shows the screen for posting new internship opportunities. The applied candidate screen allows businesses to view and manage student applications.

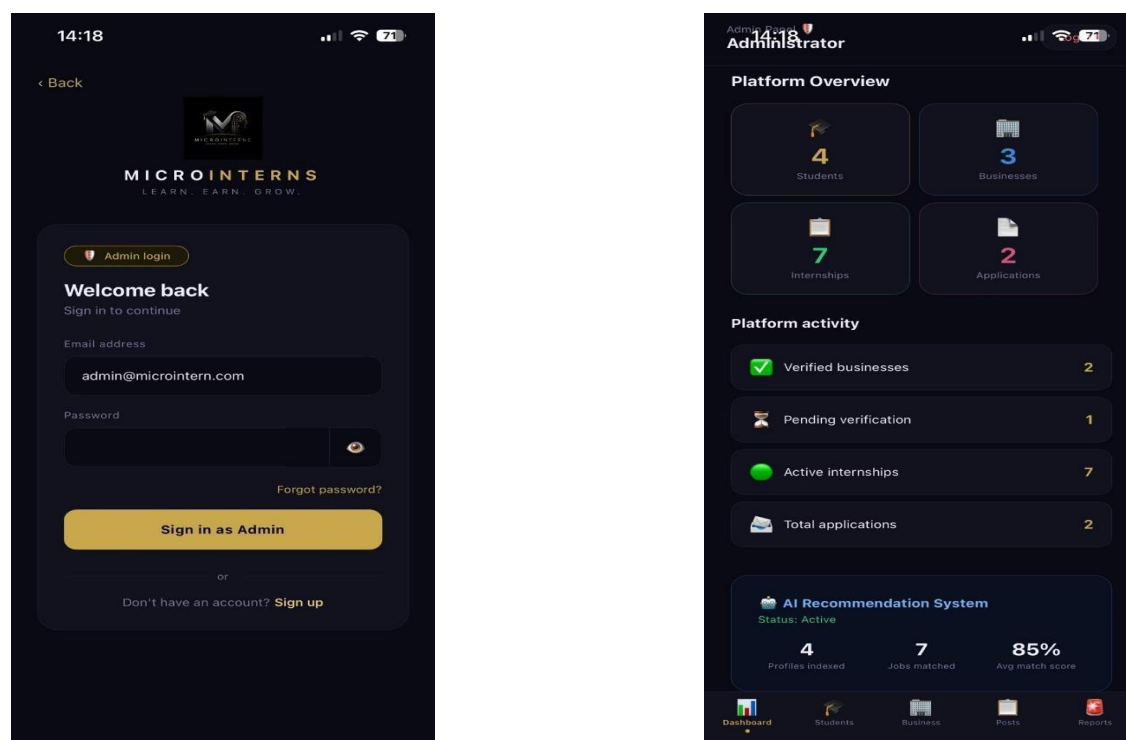


Fig 8.2.7 – Admin login & Dashboard Screen

Shows the admin login screen for secure access. The dashboard provides administrators with tools to monitor and manage the platform.

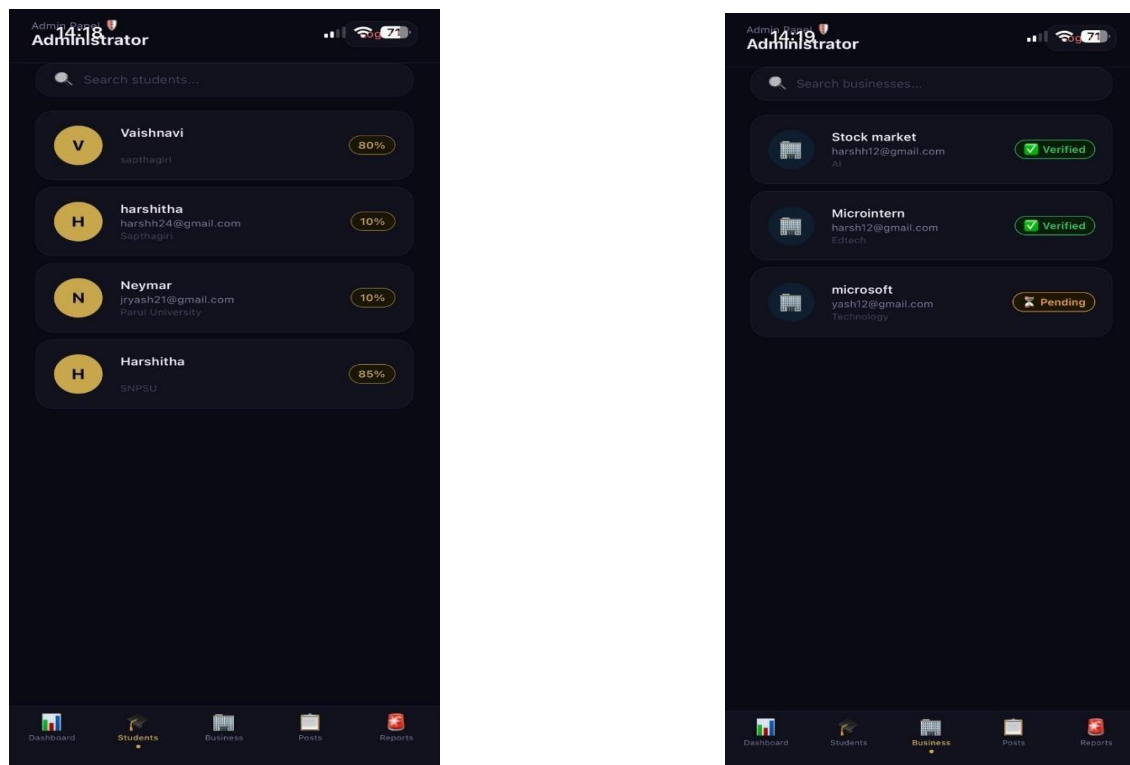


Fig 8.2.8 – View Applicants & Businesses Screen

Shows the admin screens for viewing student applicants and registered businesses. These features help administrators monitor platform activities.

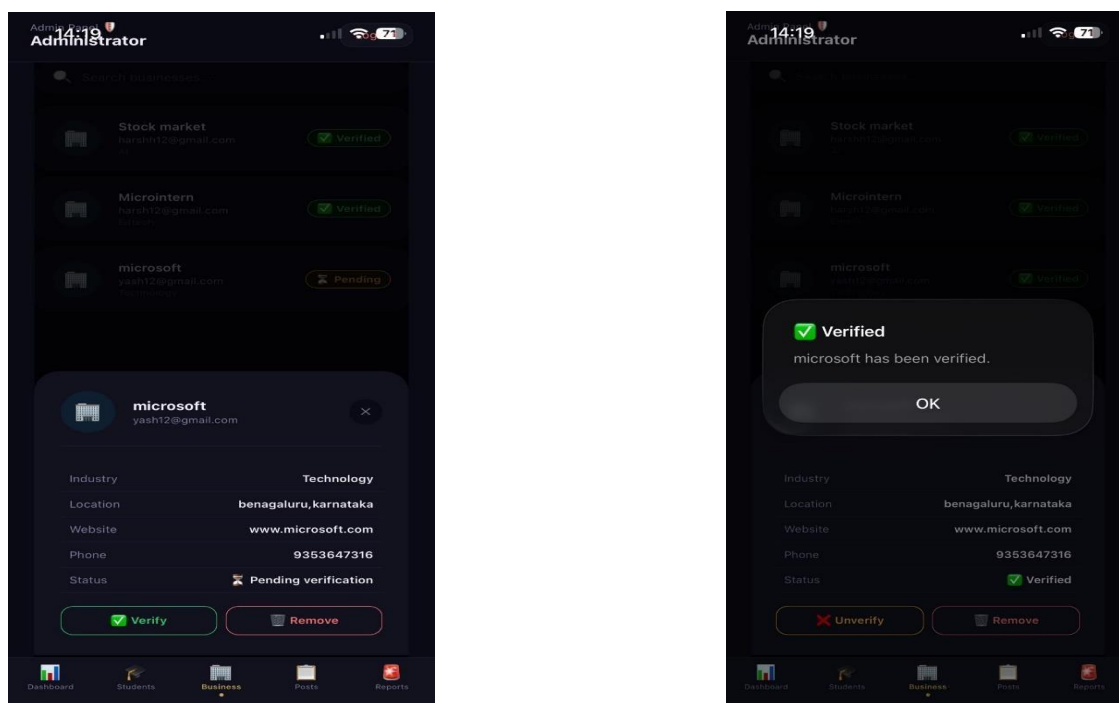


Fig 8.2.9 – Verify Business Screen

Shows the business verification screen used by the administrator. It helps verify registered businesses before allowing them to use platform services.

8.3 Reports Generated

The platform generates reports that summarize internship postings, applications, and user information in a clear tabular format. These reports help students, businesses, and administrators monitor activities and access important records efficiently.

Report Features

- Internship details
- Student and business records
- Application status
- Recently posted internships
- Application history

8.4 Performance Analysis

The application was evaluated for functionality, responsiveness, and data management. Testing confirmed that all major features operated successfully using **React Native** and **Firebase**.

Parameter	Result
User Registration	Successful
User Login	Successful
Profile Management	Successful
Resume Upload	Successful
Internship Posting	Successful
Internship Search	Successful
Internship Application	Successful
Application Status Update	Successful
Data Synchronization	Successful
Report Generation	Successful
Dashboard Loading	Fast

Table 8.4 Performance Analysis

The results indicate that the platform provides reliable performance and efficient data synchronization.

8.5 Discussion

The evaluation shows that the proposed platform successfully meets its objectives. It enables students to search and apply for internships, while businesses can post opportunities and manage applications efficiently. The integration of **React Native** and **Firebase** provides secure authentication, real-time data management, and a smooth user experience across supported mobile devices.

CHAPTER 9. CONCLUSION

9.1 Summary of Work

The **AI-Powered Micro-Internship Platform for Students and Local Businesses** was successfully designed and developed as a cross-platform mobile application using **React Native** and **Firebase**. The platform enables students to create profiles, upload resumes, search and apply for internships, while businesses can post opportunities and manage applications. Firebase Authentication, Cloud Firestore, and Firebase Storage provide secure authentication, real-time data management, and document storage. Testing confirmed that the application performs its core functions reliably and efficiently.

9.2 Contributions

The major contributions of the project include:

- Developed a cross-platform internship management application.
- Implemented secure user authentication using Firebase.
- Created separate modules for students, businesses, and administrators.
- Enabled internship posting, searching, and application tracking.
- Integrated Cloud Firestore and Firebase Storage for secure data management.
- Designed a simple and user-friendly mobile interface.

9.3 Limitations

The current version of the platform has the following limitations:

- Requires an active internet connection.
- Supports only registered users.
- Resume uploads depend on cloud storage availability.
- Provides basic internship search and filtering.
- Performance may vary with network quality.
- Advanced features such as messaging, interview scheduling, and analytics are not included.

Despite these limitations, the platform provides a secure, reliable, and effective solution for managing micro-internship opportunities between students and local businesses.

CHAPTER 10. FUTURE SCOPE

10.1 Future Enhancements

The platform can be improved by adding new features to enhance functionality and user experience. Possible enhancements include:

- Advanced internship search and filtering.
- In-app messaging between students and businesses.
- Interview scheduling and email notifications.
- Multi-language support.
- Enhanced dashboards with analytics.
- Multi-factor authentication (MFA).
- Offline support and automatic data backup.
- Integration with university placement portals and professional networking platforms.

These improvements will increase the platform's usability, security, and scalability.

10.2 Research Directions

Future research can focus on improving the platform through modern technologies and intelligent solutions. Potential research areas include:

- AI-based internship recommendation systems.
- Advanced cloud architectures for large-scale deployment.
- Biometric and multi-factor authentication.
- Blockchain-based certificate verification.
- Privacy and security for cloud recruitment systems.
- Intelligent analytics and performance optimization.

These research directions can help develop a more secure, scalable, and efficient internship management platform.

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APPENDICES

The appendices provide additional supporting information related to the development and implementation of the **AI-Powered Micro-Internship Platform for Students and Local Businesses**. They include source code, user manual, and sample data used during development and testing. These materials help in understanding the implementation details and facilitate future maintenance and enhancement of the system.

Appendix A: Source Code

App.js

```
const App = () => (
  <NavigationContainer>
    <Stack.Navigator>
      <Stack.Screen name="Login" component={Login} />
      <Stack.Screen name="Home" component={Home} />
      <Stack.Screen name="Internships" component={Internships}
    />
    <Stack.Screen name="Profile" component={Profile} />
  </Stack.Navigator>
</NavigationContainer>
);
```

FireBase Login

```
const login = async () => {
  await signInWithEmailAndPassword(auth, email, password);
  navigation.navigate("Home");
};
```

Create Internship and Get Internship

```
app.post("/internship", async (req, res) => {  
  const internship = new Internship(req.body);  
  await internship.save();  
  res.json({ message: "Internship created successfully" });  
});  
  
app.get("/internships", async (req, res) => {  
  const data = await Internship.find();  
  res.json(data);  
});
```

Apply Internship

```
app.post("/apply", async (req, res) => {  
  const application = new Application(req.body);  
  await application.save();  
  res.json({ message: "Application submitted successfully" });  
});
```

Logout

```
const logout = async () => {  
  await signOut(auth);  
  navigation.replace("Login");  
};
```

Appendix B: User Manual

The User Manual provides step-by-step instructions for operating the application.

Step 1: Login

- Enter registered email and password.
- Click Login.

Step 2: Student Registration

- Enter personal, academic, and contact details.
- Create credentials and click Register.

Step 3: Business Registration

- Enter business details and create login credentials.
- Click Register.

Step 4: Update Profile

- Add or edit personal, education, skills, or business details.
- Save changes.

Step 5: Upload Resume

- Select the resume and click Upload.

Step 6: Search Internships

- Browse available internships and view their details.

Step 7: Apply for Internship

- Select an internship and click Apply.
- Submit the application with your resume.

Step 8: Manage Internships (Business)

- Create, edit, or delete internship postings.
- Review and manage student applications.

Step 9: View Dashboard

- View internship and application information.

Step 10: Logout

- Click Logout to securely exit the application.

Appendix C: Sample Data

The application stores internship and user information using **Cloud Fire store**.

Sample Internship Data Structure

Internship Title	Company Name	Location	Duration	Required Skills	Status
Android App Developer	ABC Technologies	Bengaluru	2 Months	React Native, JavaScript	Open
UI/UX Design Intern	Creative Studio	Mysuru	1 Month	Figma, UI Design	Open
Web Development Intern	Tech Solutions	Mangaluru	3 Months	HTML, CSS, JavaScript	Open
Marketing Intern	Growth Hub	Hubballi	2 Months	Digital Marketing	Open
Content Writing Intern	EduLearn Pvt. Ltd.	Remote	1 Month	Content Writing	Open

Table Sample Dataset

The sample data demonstrates the format used by the application to store internship information. Local businesses can publish internship opportunities with similar details, and students can browse, apply for internships, and track their application status through the platform. Cloud Firestore securely stores and synchronizes this information in real time, ensuring efficient internship management and seamless interaction between students and local businesses.



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Ai-Powered Micro-Internship Platform for students and Local Business

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